

PROJECT REPORT

On 6 WEEK TRAINING AT

National Informatics Centre

(A-Block, CGO Complex, Lodhi Road, New Delhi-110 003)

Submitted by

Himanshu Dwivedi

(11/ICS/081)

GUATAM BUDDHA UNIVERSITY

SUBMITTED TO:

INTERNSHIP CERTIFICATE



भारत सरकार
संचार एवं सूचना प्रौद्योगिकी मंत्रालय
सूचना प्रौद्योगिकी विभाग
राष्ट्रीय सूचना-विज्ञान केन्द्र
ए-ब्लॉक, केन्द्रीय कार्यालय परिसर, लोधी रोड
नई दिल्ली-110003

GOVERNMENT OF INDIA
MINISTRY OF COMMUNICATIONS AND INFORMATION TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY
NATIONAL INFORMATICS CENTRE
A-BLOCK, C.G.O. COMPLEX, LODHI ROAD
NEW DELHI-110 003

ग्राम्स/GRAMS : NICNET HQ

फैक्स/FAX :

ई-मेल/E-mail :

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Himanshu Dwivedi** student of integrated M.Tech (Computer Science Engineering) at **Gautam Buddha University, Greater Noida** has successfully completed 6 week industrial training(1st June 2015 – 10th July 2015) at **National Informatics Centre**.

He worked on online cart and costing feature of the National Cloud Portal.

The project work has not been submitted to any other institution or university for the award of any degree to the best of my knowledge. He is permitted to submit report on this work as part of his industrial training.

We wish him best for the future.

हेमन्त चोपड़ा / Hemant Chopra
सहायक - सी / Saeant - B
भारत सरकार / Govt. of India
रा. सू. वि. के.ए. / N. I. C.
संचार एवं सूचना प्रौद्योगिकी मंत्रालय / Mo.C & IT
इलेक्ट्रॉनिक्स एवं सूचना प्रौद्योगिकी विभाग / DeitY
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Hemant chopra
15/7/15

Hemant Chopra
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ACKNOWLEDGMENT

I am highly grateful to the Dr. Vidushi Sharma, HOD C.S.E, Gautam Buddha University, Greater Noida, for providing this opportunity to carry out the Eight week industrial training at Spice global (at sec 125 Noida).

I would like to express my gratitude to other faculty members of Computer Science & Engineering department of ICT, for providing academic inputs, guidance & encouragement throughout the training period.

The author would like to express a deep sense of gratitude and thank to Saket Aggarwal, CEO of Spice Labs, without whose permission, wise counsel and able guidance, it would have not been possible to pursue my training in this manner.

Finally, I express my indebtedness to all who have directly or indirectly contributed to the successful completion of my industrial training.

Himanshu Dwivedi

(11/ics/081)
School of ICT
GUATAM BUDDHA UNIVERSITY

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ABOUT GAUTAM BUDDHA UNIVERSITY



GAUTAM BUDDHA UNIVERSITY

Website: www.gbu.ac.in

Address: Greater Noida, Gautam Buddha Nagar, Yamuna Express Way, Uttar Pradesh-201308 INDIA

MESSAGE FROM VICE CHANCELLOR'S DESK:

"The approach of our programmes is secular, egalitarian, and critical, relying on the Buddhist philosophy and its applicability in restoring supremacy of peace, co-existence, and development in totality. With emphasis on the rigorous training and education of our young aspirants, we ensure experiential, experimental and interdisciplinary learning and research in various fields of studies. Our meticulously designed curriculum and pedagogical training programmes stay focussed on imparting all necessary knowledge required and desired for creating humane and rational society."

VISION:

"A globally acclaimed integrated academic and research institution that creates a vibrant community of intellectuals and entrepreneurs endowed with Character, Creativity, Competence and Commitment, who can inspire meaningful transformations to ensure holistic growth and development of the society."

MISSION:

The Mission of University is expressed through the following goals;

- To generate a community of scholars who can manage continuity and change.
- To seek and incorporate the best practices in teaching and learning from around the world.
- To inculcate in the learners due appreciation for ethical, ecological and economic issues of knowledge.
- To provide knowledge based scientific solutions to satisfy the need of society and industry.
- To ensure an academic environment that guarantees avenues for all historically excluded sections of the society.

About the University:

Gautam Buddha University, established by the Uttar Pradesh Act (9) of 2002, commenced its first academic session at its 511 acres lush green campus at Greater Noida in August 2008. The University is fully funded by the New Okhla Industrial Development Authority (NOIDA) and the Greater Noida Industrial Development Authority (GNIDA), the undertakings of the Government of Uttar Pradesh. The University envisions to become a world class centre for excellence in education with a special focus to serve the under privileged and economically challenged sections of the society. The campus and is modeled as a fully residential educational campus in line with the best institutions of higher learning across the globe. The University is recognized by the University Grants Commission under UGC Act and is a member of the Association of Indian Universities.

In order to promote value-based education, research and training, the University has established eight schools of learning, which are:

- **School of Management**
- **School of Vocational Studies & Applied Sciences**
- **School of Buddhist Studies & Civilization**
- **School of Information & Communication Technology**
- **School of Biotechnology**
- **School of Humanities & Social Sciences**
- **School of Law, Justice & Governance**
- **School of Engineering**

About National Informatics Centre

National Informatics Centre

Website: www.nic.in

Address: A-Block, C.G.O. Complex, Lodhi Road, New Delhi- 110003

National Informatics Centre (NIC) was established in 1976, and has since emerged as a "prime builder" of e-Government / e-Governance applications up to the grassroots level as well as a promoter of digital opportunities for sustainable development. NIC, through its ICT Network, "NICNET", has institutional linkages with all the Ministries /Departments of the Central Government, 35 State Governments/ Union Territories, and about 625 District administrations of India. NIC has been instrumental in steering e-Government/e-Governance applications in government ministries/departments at the Centre, States, Districts and Blocks, facilitating improvement in government services, wider transparency, promoting decentralized planning and management, resulting in better efficiency and accountability to the people of India.

"Informatics-led-development" programme of the government has been spearheaded by NIC to derive competitive advantage by implementing ICT applications in social & public administration. The following major activities are being undertaken:

- Setting up of ICT Infrastructure
- Implementation of National and State Level e-Governance Projects
- Products and Services
- Consultancy to the government departments
- Research and Development
- Capacity Building

OVERVIEW OF TRAINING

In my training, I worked alongside the national cloud portal development team and worked on adding some new features in the **National Cloud Portal**.

About National Cloud Portal

NIC offers cloud services for all the government departments and ministries. National Cloud Portal is the portal from where users request for NIC Cloud Services. It has been developed using **PHP/MySQL**.

My Roles and Responsibilities:

a) Cloud Service Request and E-Cart Feature

I added an e-Cart feature where users can add various items to their carts and request for various resources on the cloud.

I worked on the database design and implementation and server side code using PHP.

b) Costing Module

I worked on a costing module for calculating the costs sheet for the services requested by the user.

I worked on the database design and implementation and server side code using PHP.

Cloud Service Request E-Cart and Costing Feature

Cloud users may request services on the cloud using the services request module. Below snapshot displays the flow for the services module:

a) Services available on cloud



b) User requests for the Web Server

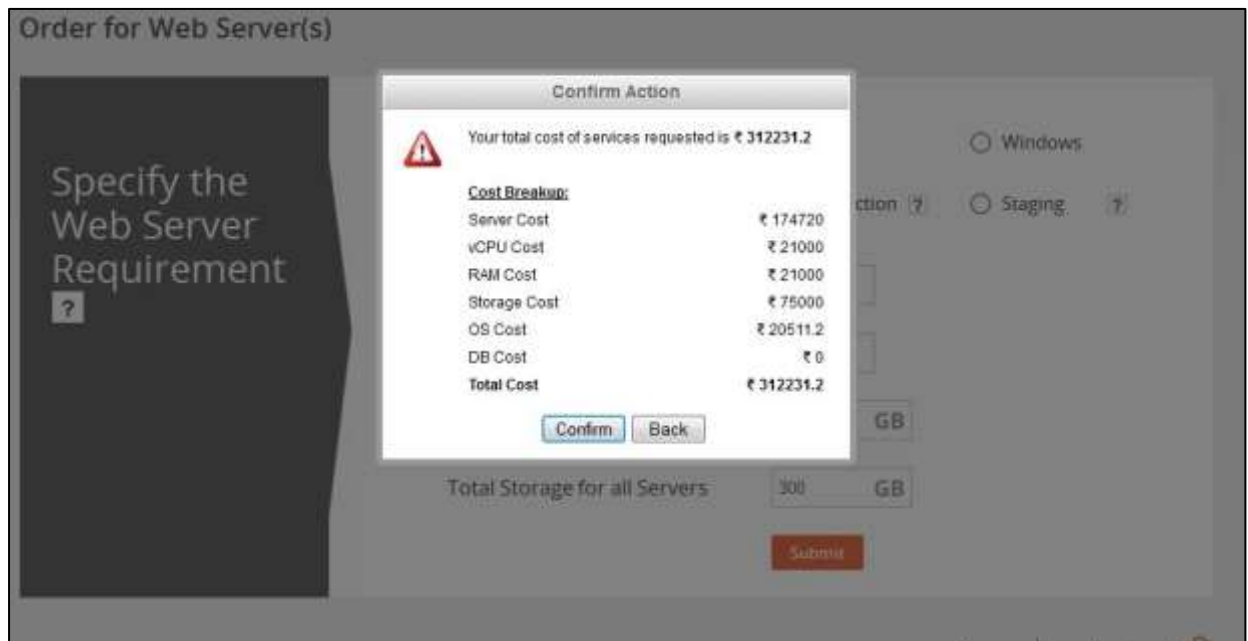
The screenshot shows the "Order for Web Server(s)" form. On the left is a dark gray sidebar with the text "Specify the Web Server Requirement" and a question mark icon. The main form area contains the following fields:

- Platform:** Radio buttons for ☒ Linux and ☐ Windows
- Environment:** Radio buttons for ☒ Production (?) and ☐ Staging (?)
- Number of Servers:** Input field with value 2
- Total vCPU for all Servers:** Input field with value 10
- Total RAM for all Servers:** Input field with value 10 and a GB unit label
- Total Storage for all Servers:** Input field with value 300 and a GB unit label
- Submit:** A red button at the bottom of the form.

At the bottom right of the interface, there is a "BACK" link and a "My Cloud Cart" icon.

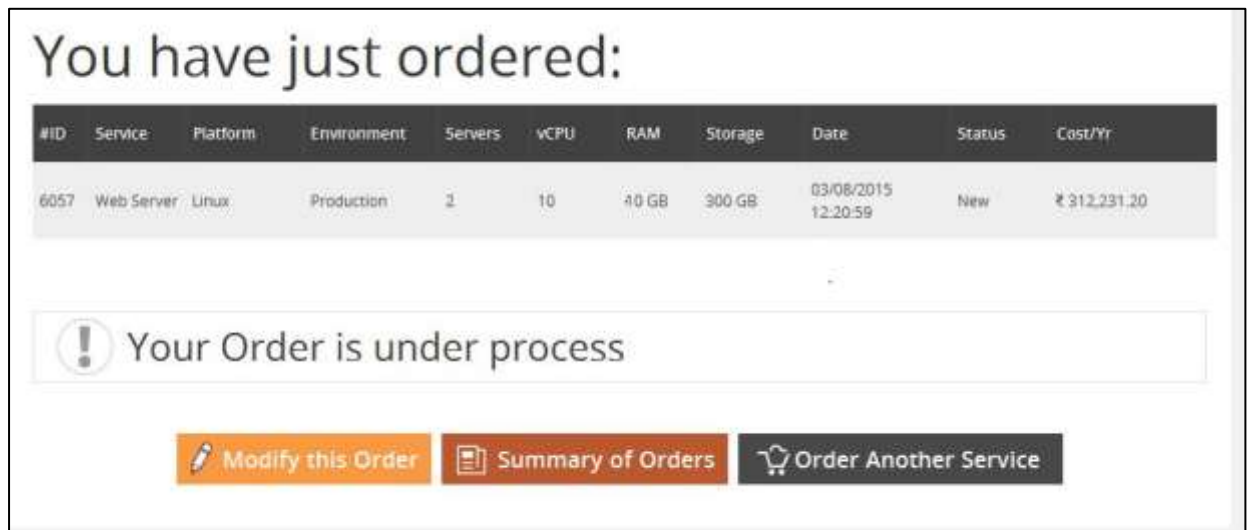
User needs to enter the various details like platform, environment and total CPU, RAM and Storage required.

c) Cost is displayed for the request to the User



Once the user confirms, the request is submitted for approval.

The order summary is then displayed to the user:



d) User submits another request, this time for an application server

Order for Application Server(s)

Specify the Application Server Requirement ?

Platform:

☒ Linux ☐ Windows

Environment:

☐ Production ? ☐ Staging ?

Number of Servers

Total vCPU for all Servers

Total RAM for all Servers

GB

Total Storage for all Servers

GB

Submit

[< BACK](#) | [My Cloud Cart](#) 

e) After submitting the requests, user may view/modify his cart.

My Cloud Cart

[ADD ANOTHER ORDER](#) 

All New Being Reviewed Approved Allocated Rejected

Total Summary (4 Orders)  

#ID	Service	Platform	Environment	Servers	vCPU	RAM	Storage	Date	Status	Cost/Yr	Action
6058	Application Server	Linux	Production	1	2	2 GB	70 GB	03/08/2015 12:22:10	New	₹ 89,540.34	 
6057	Web Server	Linux	Production	2	10	10 GB	300 GB	03/08/2015 12:20:59	New	₹ 312,231.00	 
368	Development Server	Linux And Postgres		2	2	2 GB	40 GB	08/12/2013 18:20:37	New	₹ 112,816.24	 
367	Basic Server			5	20	20 GB	400 GB	08/12/2013 18:17:43	New	₹ 501,365.00	 

CONCLUSION

In review this internship has been an excellent and rewarding experience. It was a great learning experience to work on production scale software and learn about the principles of web application design.

It gave a great exposure to the real industry environment and how to work under deadlines and how to manage things on schedule.

Skills Learnt:

Technical Skills : PHP, MySQL, SQL, SVN, WAMP

Soft Skills : Team Work, Time Management, Ownership

In essence, it was a great learning experience working with experienced developers. I got to learn a great deal about technology. This whole experience gave me a lot of exposure and broadened my perspective about how things work in industry.

